Background Reading

General resources:

- A nice paper with tips on how to organize a project. Note that you don't have to follow all these suggestions for this project, especially the section on project organization and manuscript writing are relevant: G. Wilson, J. Bryan, K. Cranston, J. Kitzes, L. Nederbragt, and T. K. Teal, “Good enough practices in scientific computing,” *PLOS Computational Biology*, (2017) **10.1371/journal.pcbi.1005510** (https://doi.org/10.1371/journal.pcbi.1005510).
- An extensive scientific writing resource: **https://www.phrasebank.manchester.ac.uk/** (https://www.phrasebank.manchester.ac.uk/)
- (https://www.phrasebank.manchester.ac.uk/) Tools for:
  - Collaborative writing: Google docs, Overleaf (Latex)
  - File sharing: Google Drive, Dropbox, **GitHub** (https://github.com/)
  - Coding: **Google Colab** (https://colab.research.google.com), **Codeshare** (https://codeshare.io/), Jupyter Notebooks, Rmarkdown Notebooks, **GitHub** (https://github.com/)
  - Collecting literature: **Zotero,** (https://www.zotero.org/) **Paperpile** (http://www.paperpile.com),

CNV data preparation

- Pre-processing: van de Wiel, M. A., Picard, F., Van Wieringen, W. N. & Ylstra, B. **Preprocessing and downstream analysis of microarray DNA copy number profiles,** (https://academic-oup-com.vu-nl.idm.oclc.org/bib/article/12/1/10/243568) *Brief. Bioinform.* (2011)
- Calling: van de Wiel MA, Kim KI, Vosse SJ, van Wieringen WN, Wilting SM, Ylstra B., **CGHcall: calling aberrations for array CGH tumor profiles** (https://academic.oup.com/bioinformatics/article/23/7/892/217759). *Bioinformatics* (2007)
- Regioning: van de Wiel MA, Wieringen WN., **CGHregions: dimension reduction for array CGH data with minimal information loss** (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2675846/?otool=inlvulib). *Cancer Inform* (2007).
- Segmentation: Venkatraman ES, Olshen AB. **A faster circular binary segmentation algorithm for the analysis of array CGH data.** (https://academic.oup.com/bioinformatics/article/23/6/657/413480) *Bioinformatics* (2007)

Machine learning

- Cross validation (**this is a key concept for this assignment**): **Wessels et al (2005) - A protocol for building and evaluating predictors of disease state based on microarray data.** (https://canvas.vu.nl/courses/87098/files/10071943/download?wrap=1) (https://canvas.vu.nl/courses/87098/files/10071943/download?download_frd=1)
- (https://canvas.vu.nl/courses/87098/files/10071943/download?wrap=1) **A guide to machine learning for biologists** (https://canvas.vu.nl/courses/87098/files/10071806?wrap=1)

(https://canvas.vu.nl/courses/87098/files/10071806/download?download_frd=1)

- **(https://canvas.vu.nl/courses/87098/files/10071806?wrap=1)** Feature selection: **Haury et al (2011) - The Influence of Feature Selection Methods on Accuracy, Stability and Interpretability of Molecular Signatures (https://canvas.vu.nl/courses/87098/files/10071826/preview)**
- **(https://canvas.vu.nl/courses/87098/files/10071826/preview)** An example of tumor classification using expression data: **van 't Veer et al (2002) - Gene expression profiling predicts clinical outcome of breast cancer (https://canvas.vu.nl/courses/87098/files/10071841/preview)**
- **(https://canvas.vu.nl/courses/87098/files/10071841/preview)** A great text book, if you want some more background in ML: **Elements of statistical learning (https://web.stanford.edu/~hastie/ElemStatLearn/)**